Day-Ahead Dispatch of a Grid-Connected Photovoltaic–Battery–Thermal System via Dynamic Programming: A Reproducible Benchmark Study

First A. Author

***Abstract*—Rising photovoltaic (PV) penetrations depress midday net load, force curtailment, and steepen evening ramps. Battery energy storage can absorb the midday surplus and reshape the thermal schedule, but many recent dispatch studies rely on heuristic or learning-based methods whose solutions cannot be checked against an exact optimum, and published case studies often omit the parameters needed to rerun them. This paper formulates the day-ahead dispatch of a grid-connected PV–battery–thermal system as a dynamic program over a discretized state space (stored energy and previous thermal output), with a quadratic thermal cost, ramp limits, charge/discharge efficiencies, state-of-charge bounds, and a curtailment penalty. The recursion is solved exactly for the discretized problem, and every input of the 24-hour benchmark instance (585 MW peak load, 21.2% PV energy share, 240 MWh/60 MW battery) is printed in the paper. Against a no-storage baseline, the optimal schedule removes 70 MWh (3.0%) of curtailment and cuts the daily cost by 2.09%; against a representative fixed-time schedule it saves 0.66%. A grid-refinement run bounds the residual discretization error below 0.06%, a capacity sweep locates the saturation of the storage value, and the optimal policy is invariant to the curtailment penalty over 100–1000 CNY/MWh. The contribution is a fully specified, easily rerun reference optimum for PV–storage dispatch studies.**

***Index Terms*—battery energy storage, curtailment, dynamic programming, energy management, photovoltaic systems, reproducibility, unit commitment.**

I. Introduction

ELECTRIC power systems are absorbing solar photovoltaic (PV) generation at a pace that changes how the remaining fleet must operate. The modernization agenda summarized under the smart-grid banner [1] anticipated much of this: metering, communication, and controllable resources on both sides of the meter. In China, the expansion is tied to the national "dual carbon" goals and to the construction of a new power system with renewables as its main body [2]. The operational signature of deep PV penetration is familiar by now. Midday net load falls toward—and in some hours below—the minimum output of the thermal fleet, so solar output must be curtailed exactly when it is most abundant; a few hours later the same system climbs its steepest ramp of the day. The California ISO operators' "duck chart" made this shape a planning fixture [3], and the structural limits it depicts were quantified for traditional power systems by Denholm and Margolis [4] and framed as a general flexibility requirement by Kondziella and Bruckner [5].

Battery energy storage is the most direct countermeasure, and a large literature now prices and schedules it. Sioshansi *et al.* estimated the arbitrage value of storage in PJM [6]; Mohsenian-Rad optimized the bidding, scheduling, and deployment of battery systems in the California day-ahead market [7]; Fang *et al.* scheduled storage under day-ahead and real-time price uncertainty with a mean–variance criterion [8]; Khaloie and Vallée dispatch liquid-air storage across coupled electricity and LNG markets [9]. Denholm and Margolis also assessed how storage and other enabling technologies relax the penetration limits of PV [10]. The common thread is that the value of storage depends on the dispatch rule that operates it, which keeps the dispatch question central.

On the dispatch side, the workhorse formulation is unit commitment and economic dispatch, covered by the standard texts [11] and served by tight mixed-integer linear formulations such as that of Carrión and Arroyo [12]. Reviews of microgrid energy management systems group the solution approaches into exact, heuristic, and learning-based families [13]. Storage sizing through economic dispatch and pumped-storage scheduling for renewable accommodation are long-standing lines in the Chinese literature [14], [15]. A growing recent thread applies deep reinforcement learning to microgrid dispatch, for example the multi-objective interval method of Mu *et al.* [16].

Two practical gaps motivated this paper. The first is verification. Heuristic and learning-based dispatch papers report their own objective values; without an exact optimum computed on the same instance, the reader cannot tell how much cost the proposed method leaves on the table. Dynamic programming, since Bellman [17], supplies such an optimum whenever the state space is small enough to enumerate—and day-ahead single-bus instances with one storage device are exactly that. The second gap is reproducibility. Case-study parameters are often summarized incompletely, so even a willing reader cannot rerun the comparison. Another method paper closes neither gap; a benchmark whose every input is printed and whose optimum is exact closes both.

This paper makes three contributions.

1) A compact dynamic-programming formulation of day-ahead PV–battery–thermal dispatch, with quadratic thermal cost, ramp limits, charge/discharge efficiencies, state-of-charge bounds, and a curtailment penalty (Sections III and IV).

Manuscript received XXX; revised XXX. (Corresponding author: First A. Author.) *(F. A. Author is with the Department of Electrical Engineering, University Name, City, Country, e-mail: first.author@example.com.)*
0000-0000-0000-0000.

AI生成
生成时间: 2026-10-25
第 1 页
AIGC:fec2aa14-47bc-41c0-8fab-fed1688c63aa

2) An exact reference optimum for a fully specified 24-hour instance—585 MW peak load, 21.2% PV energy share, 240 MWh/60 MW battery—reported beside a no-storage baseline and a fixed-time schedule (Section V). The decomposition shows where the storage value comes from on this instance: 64% curtailment-penalty avoidance, 36% fuel substitution.
3) A sensitivity study that locates capacity saturation, establishes invariance of the optimal policy to the curtailment penalty over 100–1000 CNY/MWh, and bounds the residual discretization error below 0.06% by grid refinement (Section VI).

We are equally explicit about what the paper is not. It proposes no new algorithm and claims no scalability beyond the enumerated state space. Its intended use is as a baseline against which approximate dispatch methods can be measured, and as a template for publishing complete, rerunnable instances.

The rest of the paper is organized as follows. Section II reviews related work. Section III states the model and the reference scenarios. Section IV develops the dynamic-programming solution. Section V presents the case study, Section VI the sensitivity analysis, Section VII the limitations, and Section VIII concludes.

II. Related Work

A. Exact formulations

Unit commitment and economic dispatch have a mature exact toolbox: mixed-integer linear formulations with tight relaxations [12] and the classical coordination methods collected in the standard texts [11]. Dynamic programming occupies a special place in this toolbox. Since Bellman [17], the principle of optimality has guaranteed a global optimum for sequential decision problems whose state space can be enumerated, at a runtime that is deterministic and known in advance. The cost is the curse of dimensionality: the state space grows quickly with the number of units, storage devices, and the resolution of the discretization. For single-bus day-ahead instances with one storage device and one aggregated thermal unit, enumeration is cheap, which makes DP a natural referee for approximate methods on such instances.

B. Storage valuation and scheduling

A second body of work asks what storage is worth and how it should be scheduled. Sioshansi *et al.* measured arbitrage and welfare effects in PJM [6]; Mohsenian-Rad co-optimized bidding, scheduling, and deployment in the California day-ahead market [7]; Fang *et al.* handled price uncertainty with a mean–variance criterion [8]; Khaloie and Vallée extended day-ahead dispatch to liquid-air storage coupled with LNG regasification [9]. Denholm and Margolis connected storage to the PV penetration limits of traditional systems [4], [10]. These studies assume a dispatch or bidding rule; the quality of that rule is precisely what an exact optimum calibrates.

C. Heuristic and learning-based dispatch

Reviews of microgrid energy management group solution methods into exact, heuristic, and learning-based families [13]. Deep reinforcement learning has attracted the most attention recently; Mu *et al.* perform multi-objective interval dispatch of a microgrid with deep reinforcement learning [16]. Chinese practice adds dispatch-oriented storage sizing [14] and pumped-storage scheduling for renewable accommodation [15]. Heuristic and learned policies are attractive when instances are large or uncertain, but their reported objective values are usually the only evidence of their quality. For small deterministic instances, that evidence is checkable—and rarely checked.

D. Reproducibility

Combinatorial optimization has a long culture of curated benchmark instances with certified optima. Dispatch studies with storage, by contrast, typically report results on privately specified cases. Compared with the volume of method papers, the number of published dispatch instances that a reader can rerun end-to-end is small. This paper follows the benchmark culture: the complete instance appears in Tables I and II, the solution method is a standard DP with no tunable parameters, and the optimum is reported to the yuan.

III. System Model and Problem Formulation

A. Components and notation

The system is a single bus serving a load L_t over $T = 24$ hours, with hourly resolution $\Delta t = 1$ h. Three resources meet the load.

1) Thermal unit: An aggregated thermal unit with quadratic cost

$$C(p) = ap^2 + bp + c, \tag{1}$$

output limits $p^{\min} \le p_t^{\mathrm{th}} \le p^{\max}$, and ramp limits $\left|p_t^{\mathrm{th}} - p_{t-1}^{\mathrm{th}}\right| \le R$ for $t \ge 1$. The first hour is not ramp-constrained. With the parameters of Table I, the marginal cost $2ap + b$ varies only between 281.2 and 283.6 CNY/MWh across the operating range, and the quadratic term contributes at most 0.62% of the fuel cost at full output; the unit is close to, but not exactly, a linear cost source.

2) Photovoltaic plant: A PV plant with day-ahead forecast $\bar{p}_t^{\mathrm{pv}}$. Utilized output $p_t^{\mathrm{pv}} \ge 0$ may fall below the forecast; the difference $p_t^{\mathrm{cur}} = \bar{p}_t^{\mathrm{pv}} - p_t^{\mathrm{pv}} \ge 0$ is curtailed energy, priced at λ_e CNY/MWh.

3) Battery energy storage: A battery with energy rating E_N MWh, power rating $\bar{p}^{\mathrm{ch}} = \bar{p}^{\mathrm{dis}}$ MW in both directions, charging and discharging efficiencies η^c, η^d, and stored energy s_t bounded by $s^{\min} \le s_t \le s^{\max}$ with a given initial level s_0. Charging and discharging are mutually exclusive in each hour.

B. Day-ahead dispatch problem

The power balance is

$$p_t^{\mathrm{th}} + p_t^{\mathrm{pv}} + p_t^{\mathrm{dis}} = L_t + p_t^{\mathrm{ch}}, \qquad t = 0, \ldots, T-1. \tag{2}$$

AI生成
生成时间: 2026-10-25

AIGC:fec2aa14-47bc-41c0-8fab-fed1688c63aa

The battery state evolves as

$$s_{t+1} = s_t + \eta^c p_t^{\mathrm{ch}} \Delta t - \frac{p_t^{\mathrm{dis}}}{\eta^d} \Delta t, \qquad s_0 \text{ given.} \tag{3}$$

The day-ahead problem is

$$\min \sum_{t=0}^{T-1} \Big[C(p_t^{\mathrm{th}}) + \lambda_e \, p_t^{\mathrm{cur}} \Big] + \alpha_T \left(s_T - s_0 \right)^2, \tag{4}$$

subject to (2)–(3), the output, ramp, power, and state-of-charge bounds, and mutual exclusivity of charging and discharging. The terminal penalty $\alpha_T(s_T - s_0)^2$ is a bookkeeping device: it makes the day self-contained by returning the battery to its initial level, so the reported cost is not flattered by consuming stored energy. It is not a valuation of the residual energy. Feasibility is unconditionally assured by curtailment: thermal can always back down toward $p^{\min}$ while PV output is reduced.

Two remarks on the objective. First, there is no degradation term; cycling cost is not priced, and Section VII returns to this. Second, λ_e stands in for whichever penalty or lost revenue a real operator assigns to curtailed solar; its influence is examined in Section VI.

C. Reference scenarios

Three scenarios share the instance data and the model above.

- **S1 (no storage):** $p_t^{\mathrm{ch}} = p_t^{\mathrm{dis}} = 0$ for all t. Thermal output follows net load at its minimum feasible value, subject to output and ramp limits; surplus PV above that value is curtailed.
- **S2 (fixed-time schedule):** a representative rule-based policy: charge at 25 MW in hours 10–13 (10:00–14:00), discharge at 45 MW in hours 19–21 (19:00–22:00), and recharge at 20 MW in hours 22–23 (22:00–24:00), clipped by the state-of-charge bounds. This is the kind of schedule an operator without optimization support would write by hand.
- **S3 (optimal dispatch):** the global optimum of (4) computed by dynamic programming.

IV. Dynamic Programming Solution

A. State, action, and recursion

The state after hour t is the pair $x_t = (s_t, p_{t-1}^{\mathrm{th}})$: the stored energy and the thermal output of the preceding hour, the latter carrying the ramp constraint forward. The action in hour t is the storage pair $(p_t^{\mathrm{ch}}, p_t^{\mathrm{dis}})$. Given a state and an action, the thermal output is the minimum feasible value consistent with (2),

$$p_t^{\mathrm{th}} = \max\!\big(p^{\mathrm{lo}},\; L_t + p_t^{\mathrm{ch}} - p_t^{\mathrm{dis}} - \bar{p}_t^{\mathrm{pv}}\big), \tag{5}$$

where p^{lo} collects the output floor and the ramp lower bound; utilized PV is the remainder $p_t^{\mathrm{pv}} = L_t + p_t^{\mathrm{ch}} - p_t^{\mathrm{dis}} - p_t^{\mathrm{th}}$, curtailed energy the surplus. Because C is strictly increasing, the minimum feasible thermal output is optimal for the current hour given the action; Section VII notes the one situation this restriction excludes.

Let $V_t(s, p)$ be the minimum cost-to-go from state (s, p) at time t. The recursion reads

$$V_{t+1}(s', p') = \min_{(s, p, p^{\mathrm{ch}}, p^{\mathrm{dis}})} \Big\{ V_t(s, p) + C(p^{\mathrm{th}}) + \lambda_e p^{\mathrm{cur}} \Big\}, \tag{6}$$

minimized over predecessor states and storage actions whose successor state is (s', p'), with $V_0(s_0, p) = 0$ for every thermal level (the first hour is unconstrained by ramping). The optimal value of (4) is

$$J^* = \min_{s,p} \; V_T(s, p) + \alpha_T (s - s_0)^2, \tag{7}$$

and the optimal schedule is recovered by backtracking.

B. Discretization

The state space is enumerated on grids: stored energy with step $\Delta s = 4$ MWh over $[s^{\min}, s^{\max}]$ (49 levels), thermal output with step $\Delta p = 5$ MW over $[p^{\min}, p^{\max}]$ (81 levels), and storage actions with step 10 MW in each direction (13 admissible pairs, mutual exclusivity included). A successor stored energy is snapped to the nearest grid point; the stage cost is evaluated at the actual thermal output, while the successor ramp state is the grid point nearest that output. Snapping at $\pm \Delta s/2$ is the only approximation in the recursion, and Section VI bounds its effect by grid refinement.

C. Complexity and implementation

The transition count is $T \cdot |S| \cdot |P| \cdot |A| \approx 1.2 \times 10^6$ for the benchmark grids; a solve takes about 0.2 s in pure Python (standard library only, roughly 240 lines including the three scenarios). The reference implementation is deterministic: repeated runs return bit-identical schedules and costs. The refined grid of Section VI ($\Delta s = 2$ MWh, $\Delta p = 2.5$ MW, actions of 5 MW) raises the count to 9.4×10^6 and the runtime to about 1.5 s.

V. Case Study

A. Instance

Table I lists the parameters and Table II the complete 24-hour profile; Fig. 1 plots the latter. The day peaks at 585 MW in hour 19. PV output peaks at 340 MW in hour 12, i.e., 58.1% of the load peak, and delivers 2330 MWh over the day, a 21.2% share of the 11 005 MWh load energy. The midday net-load minimum is 165 MW in hour 13, which lies 35 MW *below* the thermal floor of 200 MW: with minimum thermal output and no storage, PV cannot be fully accommodated in the midday hours, whatever the dispatch rule. This is the structural cause of curtailment in S1, and it is visible as the dip of the net-load curve in Fig. 1. The curtailment penalty is set at $\lambda_e = 500$ CNY/MWh, roughly 1.8 times the top marginal fuel cost.

B. Scenario results

Table III reports the three scenarios and Figs. 2 and 3 the schedules.

AI生成
生成时间: 2026-10-25
第 3 页
AIGC:fec2aa14-47bc-41c0-8fab-fed1688c63aa

TABLE I
BENCHMARK PARAMETERS

Subsystem	Quantity	Value
Thermal	output limits $p^{\min}/p^{\max}$	200 / 600 MW
	ramp limit R	±100 MW/h
	cost a, b, c	0.003 CNY/(MW^{2}h), 280 CNY/MWh, 5400 CNY/h
Battery	energy rating E_N	240 MWh
	power rating	60 MW (both directions)
	efficiencies η^c, η^d	0.95, 0.95
	SOC bounds / initial s_0	24–216 MWh / 120 MWh
Penalties	curtailment λ_e	500 CNY/MWh
	terminal α_T	2000 CNY/MWh^2
Instance	horizon T / resolution Δt	24 h / 1 h
	load / PV energy	11 005 / 2330 MWh

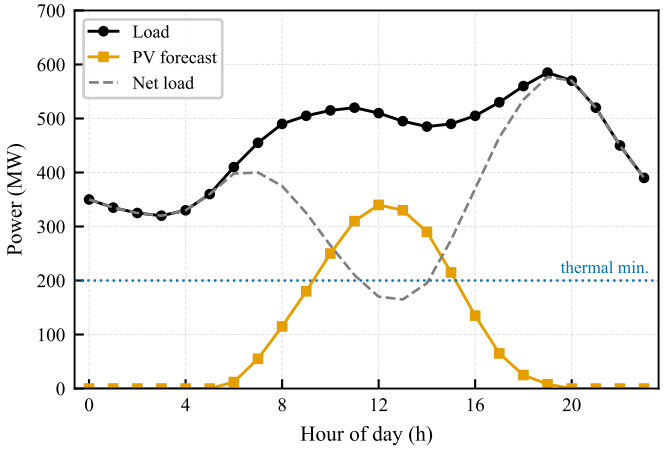


Fig. 1. Benchmark day: load, PV forecast, and net load. The midday net-load minimum (165 MW in hour 13) falls below the 200 MW thermal minimum (dotted), which is the structural cause of curtailment without storage.

1) S1: Without storage, 70 MWh of PV—3.0% of the day's solar energy—is curtailed, concentrated in hours 12–14. Total cost is 2.6236×10^6 CNY, of which the curtailment penalty is 35 000 CNY. The average specific cost of serving load is 238.4 CNY/MWh.

2) S2: The hand-written schedule absorbs part of the surplus: curtailment falls to 20 MWh (0.86%) and total cost to 2.5858×10^6 CNY, a 1.44% saving over S1. The schedule also delivers the deepest peak shaving of the three scenarios, holding thermal output at 535 MW against 577 MW in S1.

3) S3: The optimal schedule eliminates curtailment entirely and costs 2.5687×10^6 CNY: 2.09% below S1 and 0.66% below S2. Its shape (Fig. 2) is not the shape an intuition trained on peak shaving would predict. Charging runs at 40 MW through hours 11–14, lifting thermal output above its floor precisely when PV is most abundant; a sustained 20 MW discharge spans the evening hours 17–22; smaller overnight discharges (hours 0 and 5–8) and a final recharge (hour 23) complete the cycle. The battery charges and discharges 220 MWh each way, 0.92 capacity cycles, within SOC bounds of 40–200 MWh, and returns to s_0. Notably, S3 shaves the thermal peak less than S2 does (557 MW against 535 MW) at a lower total cost: given these cost parameters, storage is more valuable as an energy shifter than as a peak shaver.

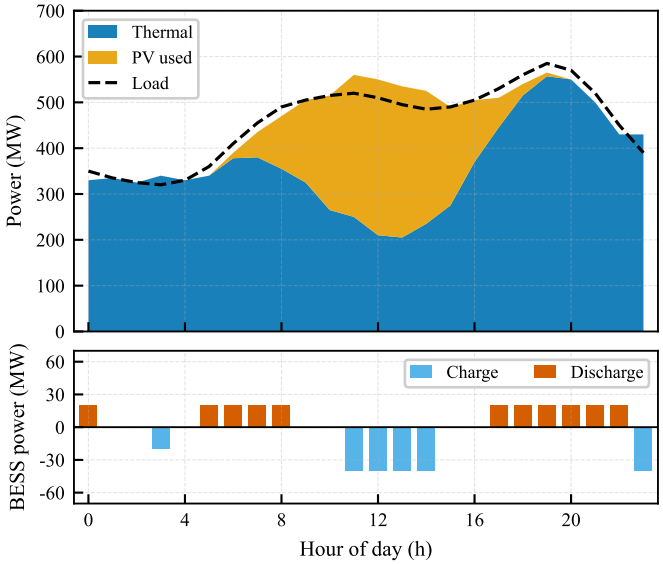


Fig. 2. Optimal dispatch (S3). Top: thermal output and utilized PV against the load curve. Bottom: battery charging (negative) and discharging (positive).

C. Where the value comes from

The 54 930 CNY saving of S3 over S1 decomposes into 35 000 CNY of avoided curtailment penalty (63.7%) and 19 930 CNY of avoided fuel (36.3%). The fuel component is almost pure energy substitution: the 70 MWh of PV that S3 accommodates displaces thermal energy at the near-constant marginal cost of 281–284 CNY/MWh, and $70 \times 282 \approx 19\,700$ CNY accounts for nearly all of it. Convexity-driven arbitrage—flattening the thermal trajectory to profit from the quadratic term in (1)—contributes little, because the marginal cost varies by only 2.4 CNY/MWh across the operating range. The corresponding decomposition of S3 over S2 is 10 000 CNY penalty and 7 076 CNY fuel.

This decomposition is a property of the instance, and it is worth stating plainly. On a day with a midday surplus and a nearly linear thermal cost, the storage value is dominated by curtailment avoidance; if a system has no penalty on curtailment and a flat marginal cost, the same battery would be worth roughly a third as much. Benchmark conclusions of this kind are exactly what an exact optimum makes possible.

VI. SENSITIVITY ANALYSIS

A. Energy capacity

Fig. 4(a) sweeps the energy rating from 60 to 480 MWh at the fixed 60 MW power rating, scaling the SOC bounds and initial level proportionally, with $\lambda_e = 500$ CNY/MWh. Three regimes are visible. Below roughly 120 MWh, capacity is scarce: at 60 MWh the optimum still curtails 20 MWh and saves 1.39%. At 120 MWh curtailment reaches zero and the saving is 1.98%; between 120 and 150 MWh the curve is flat, because the remaining value, deeper cycling, is bounded by the power rating. From 180 MWh onward the saving

AI生成
生成时间: 2026-10-25

AIGC:fec2aa14-47bc-41c0-8fab-fed1688c63aa

TABLE II
COMPLETE 24-HOUR BENCHMARK PROFILE (LOAD AND PV FORECAST)

Hour	Load (MW)	PV (MW)	Hour	Load (MW)	PV (MW)
0	350	0	12	510	340
1	335	0	13	495	330
2	325	0	14	485	290
3	320	0	15	490	215
4	330	0	16	505	135
5	360	0	17	530	65
6	410	12	18	560	25
7	455	55	19	585	8
8	490	115	20	570	0
9	505	180	21	520	0
10	515	250	22	450	0
11	520	310	23	390	0

TABLE III
SCENARIO RESULTS ON THE BENCHMARK DAY

Scenario	**Fuel** (10^6 CNY)	**Curtail. pen.** (10^6 CNY)	**Total** (10^6 CNY)	**Curtailed** (MWh)	**Curtail. rate** (%)	**Peak** p^{th} (MW)	**Saving vs. S1** (%)
S1: no storage	2.5886	0.0350	2.6236	70	3.00	577	—
S2: fixed schedule	2.5758	0.0100	2.5858	20	0.86	535	1.44
S3: DP optimum	2.5687	0.0000	2.5687	0	0.00	557	2.09

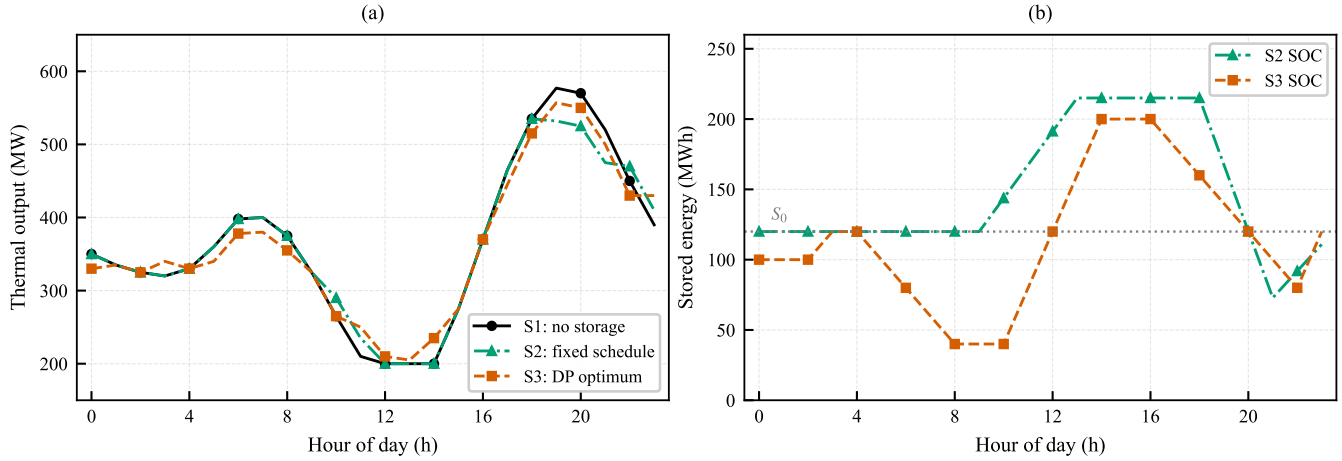


Fig. 3. Scenario comparison. (a) Thermal output. (b) Stored-energy trajectories of S2 and S3 with the initial level s_0 (dotted).

saturates at 2.09%: every additional 60 MWh of capacity beyond 180 MWh changes the total saving by less than 0.01 percentage points, and the totals at 420 and 480 MWh are identical. The reference capacity of 240 MWh lies on the flat part of the curve. For this instance there is no economic case, under either curtailment or fuel objectives, for oversizing beyond roughly three hours at rated power.

B. Curtailment penalty

Fig. 4(b) sweeps λ_e from 100 to 1000 CNY/MWh at the reference capacity. The optimal schedule and its cost are unchanged at every value tested: the optimum curtails nothing, so λ_e drops out of its objective entirely, and the saving grows linearly with λ_e merely because the no-storage baseline curtails 70 MWh whatever the penalty. Two practical readings follow. First, the policy is robust: an operator who mis-states the penalty by a factor of two deploys the same schedule. Second, the *reported* saving is sensitive to the penalty even though the policy is not—comparisons across papers that assume different λ_e are comparisons of baselines, not of methods. At $\lambda_e = 100$ CNY/MWh, below the marginal fuel cost, storage still pays 1.04% through fuel substitution alone.

C. Grid refinement

Halving all three discretization steps (2 MWh, 2.5 MW, 5 MW) changes the optimal value from 2.5687×10^6 to 2.5701×10^6 CNY, an increase of 0.055%. The sign of the change is a reminder that snapping cuts both ways: rounding the successor SOC to a coarse grid can credit energy as well as debit it, and the benchmark grid happens to sit slightly below the refined value. The refined solve still curtails nothing and still returns the battery to s_0; the scenario ranking and every qualitative conclusion are unchanged. We regard 0.06% as the

AI生成
生成时间: 2026-10-25
第 5 页
AIGC:fec2aa14-47bc-41c0-8fab-fed1688c63aa

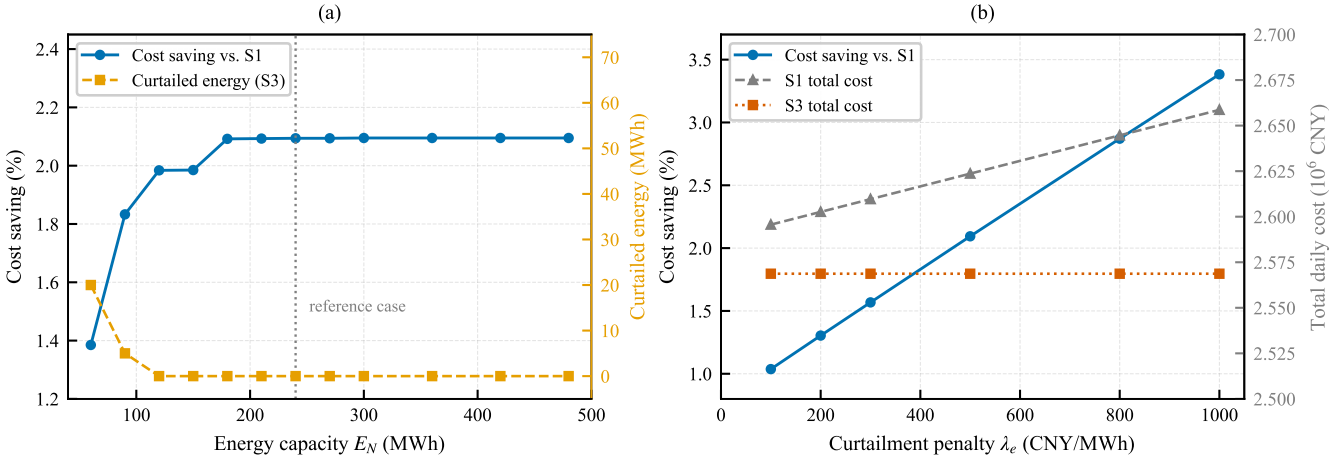


Fig. 4. Sensitivity of the storage value. (a) Capacity sweep at $\lambda_e = 500$ CNY/MWh: cost saving versus S1 (left axis) and residual curtailment under the optimal policy (right axis); the dotted line marks the reference capacity of 240 MWh. (b) Penalty sweep at $E_N = 240$ MWh: the optimal cost is invariant, while the saving grows linearly because the no-storage baseline bears $\lambda_e \times 70$ MWh.

working bound on the combined discretization artifact of the benchmark configuration.

VII. Limitations

The benchmark is deliberately small, and its boundaries should be stated as clearly as its results.

- **Single bus, single thermal unit.** There is no network, no losses, and no reliability constraint; the thermal unit is an aggregate. Multi-unit systems restore the combinatorial structure for which MILP formulations [12] or learning methods [16] are the appropriate tools. DP then serves as an offline referee on reduced instances, not as the online solver.
- **Deterministic forecasts.** PV and load are known in advance. Scheduling under uncertainty [8], [9] is outside the scope; the recursion extends to stochastic or robust variants at the usual cost in state-space size.
- **No degradation.** Cycling cost is not priced. A linear throughput cost would integrate directly into the stage cost of (6); nonlinear aging would not.
- **Discretization.** The 4 MWh/5 MW/10 MW grid and the snapping of successor states carry the artifact bounded at 0.06% in Section VI. The restriction of thermal output to the minimum feasible value in (5) excludes deliberately pre-ramping the thermal unit to reach a later high output. On this instance every net-load ramp stays within the ramp limit, so no optimal schedule is excluded; on instances with steeper load climbs the enumeration would need to admit thermal output as a decision.
- **Representative, not measured, costs.** The cost coefficients are representative values for a mid-size coal unit expressed in CNY, not measurements from a specific plant, and the profile is synthetic. The instance calibrates method comparisons; it is not evidence about any particular grid.
- **Terminal bookkeeping.** The quadratic terminal penalty restores s_0; it does not value the residual energy. Rolling horizons would price the boundary differently.

VIII. Conclusion

This paper formulated the day-ahead dispatch of a grid-connected PV–battery–thermal system as a dynamic program, solved it exactly on a discretized state space, and published the complete instance beside the optimum. On the benchmark day—585 MW peak, 21.2% PV energy share, 240 MWh/60 MW battery—the optimal schedule eliminates 70 MWh of curtailment and saves 2.09% of daily cost against no storage and 0.66% against a hand-written schedule; two thirds of the value is curtailment-penalty avoidance and one third fuel substitution. Capacity beyond roughly 180 MWh is worthless on this instance, and the optimal policy is invariant to the curtailment penalty over two orders of magnitude. The discretization artifact is bounded below 0.06%.

The intended use of these numbers is as a floor, not a ceiling: any approximate dispatch method—heuristic, learning-based, or relaxation-based—can now be measured against a certified optimum on an instance whose every input is printed above. Extending the benchmark to stochastic forecasts, degradation costs, and multi-zone networks, with the same discipline of complete disclosure, is the natural next step.

Data Availability

All instance data appear in Tables I and II. The reference implementation (Python 3, standard library only, about 240 lines, including the three scenarios and the sensitivity sweeps) and the figure-generation scripts are available from the corresponding author upon request and will be released publicly upon acceptance.

References

[1] H. Farhangi, "The path of the smart grid," *IEEE Power and Energy Magazine*, vol. 8, no. 1, pp. 18–28, Jan. 2010.

[2] Y. Shu, Y. Zhao, L. Zhao *et al.*, "Low-carbon transition pathways of china's energy and electric power under the "dual carbon" goals," *Proceedings of the CSEE*, vol. 43, no. 5, pp. 1663–1672, 2023, (in Chinese).

AI生成
生成时间: 2026-10-25
第 6 页
AIGC:fec2aa14-47bc-41c0-8fab-fed1688c63aa

[3] P. Denholm, M. O'Connell, G. Brinkman, and J. Jorgenson, "Overgeneration from solar energy in California: A field guide to the duck chart," National Renewable Energy Laboratory, Golden, CO, USA, Tech. Rep., Nov. 2015, doi: 10.2172/1226167.
[4] P. Denholm and R. M. Margolis, "Evaluating the limits of solar photovoltaics (PV) in traditional electric power systems," *Energy Policy*, vol. 35, no. 5, pp. 2852–2861, May 2007.
[5] H. Kondziella and T. Bruckner, "Flexibility requirements of renewable energy based electricity systems – a review of research results and methodologies," *Renewable and Sustainable Energy Reviews*, vol. 53, pp. 10–22, Jan. 2016.
[6] R. Sioshansi, P. Denholm, T. Jenkin, and J. Weiss, "Estimating the value of electricity storage in PJM: Arbitrage and some welfare effects," *Energy Economics*, vol. 31, no. 2, pp. 269–277, Mar. 2009.
[7] H. Mohsenian-Rad, "Optimal bidding, scheduling, and deployment of battery systems in California day-ahead energy market," *IEEE Transactions on Power Systems*, vol. 31, no. 1, pp. 442–453, Jan. 2016.
[8] X. Fang, B.-M. Hodge, L. Bai, H. Cui, and F. Li, "Mean-variance optimization-based energy storage scheduling considering day-ahead and real-time LMP uncertainties," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 7292–7295, Nov. 2018.
[9] H. Khaloie and F. Vallée, "Day-ahead dispatch of liquid air energy storage coupled with LNG regasification in electricity and LNG markets," *IEEE Transactions on Power Systems*, vol. 39, no. 3, pp. 5177–5190, May 2024.
[10] P. Denholm and R. M. Margolis, "Evaluating the limits of solar photovoltaics (PV) in electric power systems utilizing energy storage and other enabling technologies," *Energy Policy*, vol. 35, no. 9, pp. 4424–4433, Sep. 2007.
[11] A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*, 3rd ed. Hoboken, NJ, USA: John Wiley & Sons, 2013.
[12] M. Carrion and J. M. Arroyo, "A computationally efficient mixed-integer linear formulation for the thermal unit commitment problem," *IEEE Transactions on Power Systems*, vol. 21, no. 3, pp. 1371–1378, Aug. 2006.
[13] M. F. Zia, E. Elbouchikhi, and M. Benbouzid, "Microgrids energy management systems: A critical review on methods, solutions, and prospects," *Applied Energy*, vol. 222, pp. 1033–1055, Jul. 2018.
[14] X. Yang, J. Chen, L. Zhu *et al.*, "Optimal configuration of energy storage in microgrid based on economic dispatch," *Power System Protection and Control*, vol. 41, no. 1, pp. 53–59, 2013, (in Chinese).
[15] X. Ling, C. Tang, Y. Wang *et al.*, "Optimal dispatch of pumped storage power stations for promoting wind and solar power accommodation," *China Rural Water and Hydropower*, no. 5, pp. 143–148, 2026, (in Chinese).
[16] C. Mu, Y. Shi, N. Xu, X. Wang, Z. Tang, H. Jia, and H. Geng, "Multi-objective interval optimization dispatch of microgrid via deep reinforcement learning," *IEEE Transactions on Smart Grid*, vol. 15, no. 3, pp. 2957–2970, May 2024.
[17] R. Bellman, *Dynamic Programming*. Princeton, NJ, USA: Princeton University Press, 1957.

AI生成
生成时间: 2026-10-25
第 7 页
AIGC:fec2aa14-47bc-41c0-8fab-fed1688c63aa